

System Analysis & Design Project Proposal

1. Project Title

Attendance Management System

2. Group Information

Group No.: 2

Member Names & IDs:

Name	Student ID
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3. Problem Statement

Many educational institutions still maintain student attendance manually using paper registers or spreadsheets. This process is time-consuming, prone to human errors, and makes it difficult to generate attendance reports quickly. Manual record-keeping can also result in data loss and inaccurate attendance information. Therefore, an automated attendance management system is required to improve efficiency, accuracy, and record management.

4. Proposed Solution

The proposed **Attendance Management System** is a desktop-based application developed using **Python** with **MySQL** as the database. The system will enable teachers and administrators to record, manage, and monitor student attendance digitally. It will also reduce paperwork, improve data accuracy, and provide quick access to attendance records.

5. Objectives

- To develop a digital attendance management system.

- To reduce errors in manual attendance recording.
 - To store attendance records securely in a MySQL database.
 - To generate attendance reports efficiently.
 - To save time for teachers and administrators.
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6. Stakeholders & Users

Primary Users

- Teachers

Secondary Users

- Students
- Institute Authority

System Administrator

- Admin
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7. Functional Requirements (High Level)

- User Login
 - Student Management
 - Attendance Management
 - Attendance Report Generation
 - Search Student Attendance
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8. Non-Functional Requirements

Performance

- Fast response while saving and retrieving attendance records.

Security

- Secure login using username and password.
- Authorized access to attendance data.

Usability

- Simple, user-friendly, and easy-to-use interface.

Reliability

- Accurate data storage with consistent system performance.
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9. Feasibility Analysis

Technical Feasibility

The project will be developed using Python and MySQL, which are reliable, open-source, and suitable for desktop application development.

Economic Feasibility

The project is cost-effective because it uses free and open-source software tools with minimal development expenses.

Operational Feasibility

Teachers and administrators can easily use the system after a short training session, making it practical for everyday attendance management.

10. Tools & Technologies (Proposed)

- **Programming Language:** Python
 - **Database:** MySQL
 - **Modeling Tool:** Draw.io
 - **Documentation Tool:** Microsoft Word
 - **Prototype Tool (Optional):** Figma
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11. Expected Deliverables

- Project Proposal
- Software Requirement Specification (SRS)
- UML Diagrams
- ER Diagram
- System Architecture Diagram
- Attendance Management System Software
- Final Project Report